

TestPlatform

Rapport d'Execution de Campagne

test

Projet: ms_gestion

Version: 2.5

Environnement: ms_gestion_env

Debut: 30/06/2026 10:19 | Fin: 30/06/2026 10:21

Duree: 93.8 s

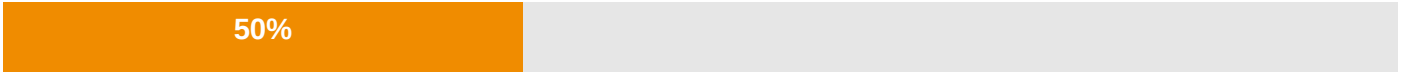
Rapport genere le 30/06/2026 10:21

PARTIEL

Resume executif

Total	Succes	Echecs	Erreurs	Duree
4	2	2	0	93.8 s

Taux de reussite



Evolution vs campagne precedente: N/A

Premiere campagne

Corrections IA: 4 test(s) corrige(s) automatiquement, 2 avec succes

Informations du contexte

Projet	ms_gestion
Description	test

Suites

Nom	Type	Depot Git	Branche
Suite UNIT	UNIT	N/A	main
Suite INTEGRATION	INTEGRATION	N/A	main

Cas de test

Titre	Type	Priorite	Risque	Script
create project	UNIT	2	HIGH	IA
create envirenemnt	INTEGRATION	5	CRITICAL	IA
List environnement	INTEGRATION	2	CRITICAL	IA
add test case	INTEGRATION	2	CRITICAL	IA

Environnement

URL Web	http://localhost:3000
URL API	N/A
Repository	https://github.com/ilyeschaabani/Automatisation_intelligente_des_tests_applicatifs
Branche	main
Base de donnees	POSTGRESQL

Resultats details

Synthese par suite

Suite	Total	Succes	Echecs	Taux
Suite UNIT	1	0	1	0%
Suite INTEGRATION	3	2	1	67%

Suite UNIT

create project		IA x3		FAILURE
Type	UNIT	Duree	11.1 s	
Script	IA	Erreur	[COMPILATION_ERROR] COMPILATION ERROR :	

Suite INTEGRATION

create envirenemnt		IA x1		SUCCESS
Type	INTEGRATION	Duree	11.0 s	
Script	IA	Erreur	[COMPILATION_ERROR] —	

List environnement		IA x2		SUCCESS
Type	INTEGRATION	Duree	24.7 s	
Script	IA	Erreur	[COMPILATION_ERROR] —	

add test case		IA x3		FAILURE
Type	INTEGRATION	Duree	24.4 s	
Script	IA	Erreur	[COMPILATION_ERROR] Tests run: 1, Failures: 1, Errors: 0, Skipped: 0, Time elapsed: 12.77 s <<< FAILURE! -- in suites.integration.TestCaseServiceIT	

Analyse IA

create project

Erreurs extraites	<pre>[ERROR] COMPILATION ERROR : [ERROR] /C:/tmp/ms-execution/exec-1126529147941670935/Backend/ms_gestion/src/test/java/suites/unit/ProjectServiceCreateHappyTest.java:33: cannot find symbol [ERROR] /C:/tmp/ms-execution/exec-1126529147941670935/Backend/ms_gestion/src/test/java/suites/unit/ProjectServiceCreateHappyTest.java:36: cannot find symbol [INFO] BUILD FAILURE [ERROR] Failed to execute goal org.apache.maven.plugins:maven-compiler-plugin:3.14.1:testCompile (default-test-compile) on project ms_gestion: Compilation failure: Compilation failure: [ERROR] /C:/tmp/ms-execution/exec-1126529147941670935/Backend/ms_gestion/src/test/java/suites/unit/ProjectServiceCreateHappyTest.java:33: cannot find symbol [ERROR] symbol: class ProjectAccessService [ERROR] location: class suites.unit.ProjectServiceCreateHappyTest [ERROR] /C:/tmp/ms-execution/exec-1126529147941670935/Backend/ms_gestion/src/test/java/suites/unit/ProjectServiceCreateHappyTest.java:36: cannot find symbol [ERROR] symbol: class ProjectService [ERROR] location: class suites.unit.ProjectServiceCreateHappyTest [ERROR] -> [Help 1] [ERROR] [ERROR] To see the full stack trace of the errors, re-run Maven with the -e switch. [ERROR] Re-run Maven using the -X switch to enable full debug logging. [ERROR] [ERROR] For more information about the errors and possible solutions, please read the following articles: [ERROR] [Help 1] http://cwiki.apache.org/confluence/display/MAVEN/MojoFailureException</pre>
Analyse IA	**Cause** : Les classes `ProjectAccessService` (ligne 33) et `ProjectService` (ligne 36) sont indiquées comme non trouvées, ce qui indique qu'elles ne sont pas dans le classpath de compilation des tests (probablement absentes du projet ou mal structurées). **Correction** : Vérifie que `ProjectAccessService.java` et `ProjectService.java` existent bien dans le module source (`src/main/java/com/pfe/platform/ms_gestion/service/`), qu'elles sont déclarées comme `public`, et qu'aucun erreur de compilation dans les sources ne bloque la compilation des tests. Si le projet est modulaire, assure-toi qu'elles sont exportées/visibles dans `module-info.java` ou dans les dépendances de compilation. **Conseil** : Exécute `mvn clean compile` avant `mvn test-compile` pour t'assurer que les classes sont compilées avant les tests.

create envirenemnt

Erreurs extraites	<pre>Hibernate: drop table if exists environments cascade Hibernate: drop table if exists project_members cascade Hibernate: drop table if exists projects cascade Hibernate: drop table if exists security_scans cascade Hibernate: drop table if exists security_vulnerabilities cascade Hibernate: drop table if exists test_cases cascade Hibernate: drop table if exists test_suites cascade Hibernate: create table campaign_testcases (execution_order integer, campaign_id bigint not null, test_case_id bigint not null, primary key (campaign_id, test_case_id)) Hibernate: create table campaigns (created_at timestamp(6), environment_id bigint not null, finished_at timestamp(6), id bigint generated by default as identity, project_id bigint not null, started_at timestamp(6), app_version varchar(255), git_branch varchar(255), name varchar(255) not null, status varchar(255) check ((status in ('PENDING','RUNNING','FINISHED','FINISHED_WITH_ERRORS','ABORTED'))), trigger_mode</pre>
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```
varchar(255) check ((trigger_mode in ('MANUAL','SCHEDULED','CI'))), primary key (id))
Hibernate: create table compliance_results (failed_controls integer, passed_controls
integer, score float(53), total_controls integer, evaluated_at timestamp(6), id bigint
generated by default as identity, project_id bigint, framework varchar(255) not null,
version varchar(255), primary key (id))
Hibernate: create table environments (created_at timestamp(6), id bigint generated by
default as identity, project_id bigint not null, base_url_api varchar(255), base_url_web
varchar(255), database_type varchar(255), git_branch varchar(255), git_repo_url
varchar(255), name varchar(255) not null, primary key (id))
Hibernate: create table project_members (id bigint generated by default as identity,
project_id bigint not null, user_id bigint not null, primary key (id), unique (project_id,
user_id))
Hibernate: create table projects (created_at timestamp(6), created_by bigint, id bigint
generated by default as identity, description TEXT, git_default_branch varchar(255),
git_repo_url varchar(255), name varchar(255) not null unique, status varchar(255) check
((status in ('ACTIVE','PAUSED','ARCHIVED'))), primary key (id))
Hibernate: create table security_scans (coverage float(53), files_analyzed integer,
lines_analyzed integer, completed_at timestamp(6), id bigint generated by default as
identity, project_id bigint, started_at timestamp(6), branch varchar(255), commit_hash
varchar(255), duration varchar(255), engine varchar(255) not null, scan_ref varchar(255) not
null, scan_type varchar(255) not null check ((scan_type in ('SAST','DAST','SCA'))), status
varchar(255) not null check ((status in ('RUNNING','COMPLETED','FAILED','SCHEDULED'))),
target_url varchar(255), primary key (id))
Hibernate: create table security_vulnerabilities (line integer, created_at timestamp(6), id
bigint generated by default as identity, project_id bigint, scan_id bigint, updated_at
timestamp(6), assigned_to varchar(255), cwe_id varchar(255), description TEXT, endpoint
varchar(255), evidence TEXT, file varchar(255), fix_example TEXT, http_method varchar(255),
owasp_category varchar(255), parameter varchar(255), recommendation TEXT, risk TEXT,
severity varchar(255) not null check ((severity in
('CRITICAL','HIGH','MEDIUM','LOW','INFO'))), snippet TEXT, source varchar(255), status
varchar(255) not null check ((status in
('OPEN','IN_PROGRESS','RESOLVED','CLOSED','FALSE_POSITIVE'))), title varchar(255) not null,
vuln_type varchar(255) not null check ((vuln_type in ('SAST','DAST','SCA'))), primary key
(id))
Hibernate: create table test_cases (active boolean, flaky boolean, generated boolean not
null, max_duration_seconds integer, priority integer, created_at timestamp(6), id bigint
generated by default as identity, suite_id bigint not null, database_type varchar(255),
description TEXT, generated_code TEXT, git_repo_url varchar(255), risk_level varchar(255)
check ((risk_level in ('CRITICAL','HIGH','MEDIUM','LOW'))), script_path varchar(255),
spring_profile varchar(255), tags varchar(255), target_class_name varchar(255), title
varchar(255) not null, type varchar(255) check ((type in
('WEB','API','UNIT','INTEGRATION','UX_WEB','UX_MOBILE'))), test_data JSONB, primary key
(id))
Hibernate: create table test_suites (created_at timestamp(6), id bigint generated by default
as identity, project_id bigint not null, description TEXT, git_branch varchar(255),
git_repo_url varchar(255), module_path varchar(255), name varchar(255) not null, type
varchar(255) check ((type in ('WEB','API','UNIT','INTEGRATION','UX_WEB','UX_MOBILE'))),
primary key (id))
Hibernate: alter table if exists campaign_testcases add constraint
FKdyri4dtm81vciauhdyekpeh1t foreign key (campaign_id) references campaigns
Hibernate: alter table if exists campaign_testcases add constraint
FK9weg2djl986e8fnmb11onvyg6 foreign key (test_case_id) references test_cases
Hibernate: alter table if exists campaigns add constraint FKmyimryhxtshr5smm4vb6y7ny8
foreign key (environment_id) references environments
Hibernate: alter table if exists campaigns add constraint FK8tqxs91vmi30cb4pkapa432mi
foreign key (project_id) references projects
Hibernate: alter table if exists compliance_results add constraint
FKbc45lcknv6hj2rrknhtx0rsu foreign key (project_id) references projects
Hibernate: alter table if exists environments add constraint FKnlc5pk4efot15s7hyrqah6ru4
foreign key (project_id) references projects
Hibernate: alter table if exists project_members add constraint FKdki1sp2homqsdcvqm9yrix31g
foreign key (project_id) references projects
```

```
Hibernate: alter table if exists security_scans add constraint FKal1fccjhlqwsdiyfe0xgn8hag
foreign key (project_id) references projects
Hibernate: alter table if exists security_vulnerabilities add constraint
FK5dqkgk4x5b5h2qub892el4l45 foreign key (project_id) references projects
Hibernate: alter table if exists security_vulnerabilities add constraint
FK41kves1cithp1miw1j7n7v8gi foreign key (scan_id) references security_scans
Hibernate: alter table if exists test_cases add constraint FKeuc59ejswhi3pn4lv997yd82
foreign key (suite_id) references test_suites
Hibernate: alter table if exists test_suites add constraint FK9m60ulvcghbrs28iuc98ug1vb
foreign key (project_id) references projects
2026-06-29T17:02:04.714+01:00 INFO 36476 --- [ms_gestion] [          main]
j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for
persistence unit 'default'
2026-06-29T17:02:04.873+01:00 INFO 36476 --- [ms_gestion] [          main]
o.s.d.j.r.query.QueryEnhancerFactories   : Hibernate is in classpath; If applicable, HQL
parser will be used.
2026-06-29T17:02:06.471+01:00 WARN 36476 --- [ms_gestion] [          main]
JpaBaseConfiguration$JpaWebConfiguration : spring.jpa.open-in-view is enabled by default.
Therefore, database queries may be performed during view rendering. Explicitly configure
spring.jpa.open-in-view to disable this warning
2026-06-29T17:02:06.617+01:00 WARN 36476 --- [ms_gestion] [          main]
.s.a.UserDetailsServiceAutoConfiguration :
```

Using generated security password: d7b440be-d2f1-4671-a95e-e351a9d29b45

This generated password is for development use only. Your security configuration must be updated before running your application in production.

```
2026-06-29T17:02:06.630+01:00 INFO 36476 --- [ms_gestion] [          main]
r$InitializeUserDetailsServiceConfigurer : Global AuthenticationManager configured with
UserDetailsService bean with name inMemoryUserDetailsService
2026-06-29T17:02:07.533+01:00 INFO 36476 --- [ms_gestion] [          main]
s.integration.EnvironmentServiceIT       : Started EnvironmentServiceIT in 12.637 seconds
(process running for 13.746)
Hibernate: insert into projects
(created_at,created_by,description,git_default_branch,git_repo_url,name,status) values
(?,?,?,?,?,?,?)
Hibernate: insert into project_members (project_id,user_id) values (?,?)
Hibernate: select pm1_0.id from project_members pm1_0 where pm1_0.project_id=? and
pm1_0.user_id=? fetch first ? rows only
Hibernate: select e1_0.id from environments e1_0 where e1_0.project_id=? and e1_0.name=?
fetch first ? rows only
Hibernate: insert into environments
(base_url_api,base_url_web,created_at,database_type,git_branch,git_repo_url,name,project_id)
values (?,?,?,?,?,?,?)
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 14.40 s -- in
suites.integration.EnvironmentServiceIT
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 20.658 s
[INFO] Finished at: 2026-06-29T17:02:09+01:00
[INFO] -----
```

----- STDERR -----

Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended

Analyse	1. Ce qui est bien testé : Le test couvre correctement la <i>happy path</i> de la méthode <code>add</code> d' <code>EnvironmentService</code> , incluant la validation des données d'entrée, la persistance en base (via <code>EnvironmentRepository</code>), la gestion du contexte sécurité (via <code>SecurityContextHolder</code>), et la vérification de l'appartenance à un projet (<code>ProjectMember</code>). L'environnement de test H2 en mémoire et la configuration <code>@Transactional + @Rollback</code> garantissent l'isolation et le nettoyage automatique.
IA	2. Suggestion d'amélioration : Ajouter des assertions sur la structure de l'objet retourné (ex : <code>@NotBlank</code> sur <code>name</code> , <code>@URL</code> sur les <code>baseUrl</code>) via un <i>validateur</i> (ex : <code>Validator.validate(request)</code>) pour s'assurer que les contraintes Bean Validation sont bien appliquées <i>avant</i> la persistance. Cela renforcerait la robustesse du test face à d'éventuels changements de contraintes métier.

List environnement

Erreurs	Hibernate: drop table if exists projects cascade
extraites	Hibernate: drop table if exists security_scans cascade
	Hibernate: drop table if exists security_vulnerabilities cascade
	Hibernate: drop table if exists test_cases cascade
	Hibernate: drop table if exists test_suites cascade
	Hibernate: create table campaign_testcases (execution_order integer, campaign_id bigint not null, test_case_id bigint not null, primary key (campaign_id, test_case_id))
	Hibernate: create table campaigns (created_at timestamp(6), environment_id bigint not null, finished_at timestamp(6), project_id bigint not null, started_at timestamp(6), app_version varchar(255) not null, status varchar(255) check ((status in ('PENDING','RUNNING','FINISHED','FAILED'))), trigger_mode varchar(255) check ((trigger_mode in ('MANUAL','SCHEDULED','CI'))), primary key (id))
	Hibernate: create table compliance_results (failed_controls integer, passed_controls integer, score float(53), evaluated_at timestamp(6), id bigint generated by default as identity, project_id bigint, framework varchar(255), primary key (id))
	Hibernate: create table environments (created_at timestamp(6), id bigint generated by default as identity, base_url_api varchar(255), base_url_web varchar(255), database_type varchar(255), git_branch varchar(255) not null, primary key (id))
	Hibernate: create table project_members (id bigint generated by default as identity, project_id bigint, user_id bigint, primary key (id), unique (project_id, user_id))
	Hibernate: create table projects (created_at timestamp(6), created_by bigint, id bigint generated by default as identity, git_default_branch varchar(255), git_repo_url varchar(255), name varchar(255) not null unique ((name, git_repo_url)), status varchar(255) check ((status in ('ACTIVE','PAUSED','ARCHIVED'))), primary key (id))
	Hibernate: create table security_scans (coverage float(53), files_analyzed integer, lines_analyzed integer, id bigint generated by default as identity, project_id bigint, started_at timestamp(6), branch varchar(255), duration varchar(255), engine varchar(255) not null, scan_ref varchar(255) not null, scan_type varchar(255) check ((scan_type in ('SAST','DAST','SCA'))), status varchar(255) not null check ((status in ('RUNNING','COMPLETED','FAILED'))), primary key (id))
	Hibernate: create table security_vulnerabilities (line integer, created_at timestamp(6), id bigint generated by default as identity, project_id bigint, scan_id bigint, updated_at timestamp(6), assigned_to varchar(255), cwe_id varchar(255), evidence TEXT, file varchar(255), fix_example TEXT, http_method varchar(255), owa varchar(255), recommendation TEXT, risk TEXT, severity varchar(255) not null check ((severity in ('CRITICAL','HIGH','MEDIUM','LOW','INFO'))), snippet TEXT, source varchar(255), status varchar(255) check ((status in ('OPEN','IN_PROGRESS','RESOLVED','CLOSED','FALSE_POSITIVE'))), title varchar(255) not null, vuln_type varchar(255) check ((vuln_type in ('SAST','DAST','SCA'))), primary key (id))
	Hibernate: create table test_cases (active boolean, flaky boolean, generated boolean not null, id bigint generated by default as identity, suite_id bigint not null, description TEXT, generated_code TEXT, git_repo_url varchar(255), risk_level varchar(255) check ((risk_level in ('CRITICAL','HIGH','MEDIUM','LOW'))), script_path varchar(255), spring_profile varchar(255), tags varchar(255), title varchar(255) not null, type varchar(255) check ((type in ('WEB','API','UNIT','INTEGRATION','UX_WEB','UX_MOBILE'))), test_data JSONB, primary key (id))
	Hibernate: create table test_suites (created_at timestamp(6), id bigint generated by default as identity, description TEXT, git_branch varchar(255), git_repo_url varchar(255), module_path varchar(255), name varchar(255) check ((type in ('WEB','API','UNIT','INTEGRATION','UX_WEB','UX_MOBILE'))), primary key (id))
	Hibernate: alter table if exists campaign_testcases add constraint FKdyri4dtm81vciauhdyekpeh1t foreign key (campaign_id) references campaigns (id)

```
Hibernate: alter table if exists campaign_testcases add constraint FK9weg2djlq86e8fnmb11onvyg6 test_cases
Hibernate: alter table if exists campaigns add constraint FKmyimryhxtshr5smm4vb6y7ny8 foreign key
Hibernate: alter table if exists campaigns add constraint FK8tqxs91vmi30cb4pkapa432mi foreign key
Hibernate: alter table if exists compliance_results add constraint FKbc45lcknv6hj2rrknhttx0rsu projects
Hibernate: alter table if exists environments add constraint FKnc1c5pk4efot15s7hyrqah6ru4 foreign key
Hibernate: alter table if exists project_members add constraint FKdki1sp2homqsdvcvm9yrix31g foreign key
Hibernate: alter table if exists security_scans add constraint FKal1fccjhlqwsdiyfe0xgn8hag foreign key
Hibernate: alter table if exists security_vulnerabilities add constraint FK5dqkgk4x5b5h2qub892e projects
Hibernate: alter table if exists security_vulnerabilities add constraint FK41kves1citph1miw1j7n security_scans
Hibernate: alter table if exists test_cases add constraint FKeuc59ejswhi3pn4lv997yd82 foreign key
Hibernate: alter table if exists test_suites add constraint FK9m60ulvcghbrs28iuc98ug1vb foreign key
2026-06-29T17:12:04.198+01:00 INFO 34636 --- [ms_gestion] [ main] j.LocalContainerEntityManagerFactory for persistence unit 'default'
2026-06-29T17:12:04.377+01:00 INFO 34636 --- [ms_gestion] [ main] o.s.d.j.r.query.QueryExecutor classpath; If applicable, HQL parser will be used.
2026-06-29T17:12:06.407+01:00 WARN 34636 --- [ms_gestion] [ main] JpaBaseConfiguration in-view is enabled by default. Therefore, database queries may be performed during view rendering in-view to disable this warning
2026-06-29T17:12:06.570+01:00 WARN 34636 --- [ms_gestion] [ main] .s.a.UserDetailsService

Using generated security password: f371cc0f-ea23-486e-a416-20ef9b1d3b11

This generated password is for development use only. Your security configuration must be updated in production.

2026-06-29T17:12:06.582+01:00 INFO 34636 --- [ms_gestion] [ main] r$InitializeUserDetailsService
AuthenticationManager configured with UserDetailsServiceImpl bean with name inMemoryUserDetailsService
2026-06-29T17:12:07.454+01:00 INFO 34636 --- [ms_gestion] [ main] s.integration.EnvironmentServiceIT in 12.311 seconds (process running for 13.322)
Hibernate: insert into projects (created_at,created_by,description,git_default_branch,git_repo_url) values (?,?)
Hibernate: insert into project_members (project_id,user_id) values (?,?)
Hibernate: select pm1_0.id from project_members pm1_0 where pm1_0.project_id=? and pm1_0.user_id=?
Hibernate: select e1_0.id from environments e1_0 where e1_0.project_id=? and e1_0.name=? fetch size 1
Hibernate: insert into environments (base_url_api,base_url_web,created_at,database_type,git_branch) values (?, ?, ?, ?, ?, ?)
Hibernate: select pm1_0.id from project_members pm1_0 where pm1_0.project_id=? and pm1_0.user_id=?
Hibernate: select e1_0.id,e1_0.base_url_api,e1_0.base_url_web,e1_0.created_at,e1_0.database_type,e1_0.git_branch,e1_0.name
from environments e1_0 where e1_0.project_id=?
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 14.01 s -- in suites.integration
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 20.202 s
[INFO] Finished at: 2026-06-29T17:12:09+01:00
[INFO] -----

----- STDERR -----
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
```

Analyse
IA

****1. Ce qui est bien testé (1-2 lignes)****
Le test valide correctement le flux complet *happy path* de création et de listage d'un environnement, la persistance en BDD (via les repositories injectés), la transformation en DTO (`EnvironmentRepository`), le contexte utilisateur (via `SecurityContextHolder`). Les assertions couvrent l'intégralité des cas de test.

****2. Suggestion d'amélioration (1-2 lignes)****

Pour renforcer la ****robustesse****, il serait préférable de tester explicitement l'****idempotence**** avec le même nom/Git repo URL existe déjà, le service doit gérer le conflit – via exception ou séparer la création et la liste en deux tests distincts améliorerait la ****lisibilité et le diagnostic (SURPRISE)****).

add test case

Erreurs extraites	<pre>[ERROR] Tests run: 1, Failures: 1, Errors: 0, Skipped: 0, Time elapsed: 12.77 s <<< FAILURE! -- in suites.integration.TestCaseServiceIT [ERROR] suites.integration.TestCaseServiceIT.test_add_happy_path -- Time elapsed: 0.195 s <<< FAILURE! java.lang.AssertionError: expected [src/test/java/com/example/user/AddUserTest.java] but found [null] [ERROR] Failures: [ERROR] TestCaseServiceIT>AbstractTestNGSpringContextTests.run:153->test_add_happy_path:122 expected [src/test/java/com/example/user/AddUserTest.java] but found [null] [ERROR] Tests run: 1, Failures: 1, Errors: 0, Skipped: 0 [INFO] BUILD FAILURE [ERROR] Failed to execute goal org.apache.maven.plugins:maven-surefire-plugin:3.2.5:test (default-test) on project ms_gestion: There are test failures. [ERROR] [ERROR] Please refer to C:\tmp\ms-execution\exec-13892590320815266371\Backend\ms_gestion\target\surefire-reports for the individual test results. [ERROR] Please refer to dump files (if any exist) [date].dump, [date]-jvmRun[N].dump and [date].dumpstream. [ERROR] -> [Help 1] [ERROR] [ERROR] To see the full stack trace of the errors, re-run Maven with the -e switch. [ERROR] Re-run Maven using the -X switch to enable full debug logging. [ERROR] [ERROR] For more information about the errors and possible solutions, please read the following articles: [ERROR] [Help 1] http://cwiki.apache.org/confluence/display/MAVEN/MojoFailureException</pre>
Analyse IA	**Cause** : La méthode <code>testCaseService.add(suiteId, request)</code> renvoie un <code>TestCaseResponse</code> dont le champ <code>scriptPath</code> est <code>null</code>, alors que le test s'attend à ce qu'il soit égal à <code>"src/test/java/com/example/user/AddUserTest.java"</code> (lignes 117-122 dans <code>test_add_happy_path</code>). Cela indique que le service ou la couche de persistance ne transfère pas correctement ce champ depuis la requête vers l'entité ni vers la réponse. **Correction** : Vérifie que <code>TestCaseService.add(...)</code> mappe explicitement le champ <code>scriptPath</code> de <code>CreateTestCaseRequest</code> vers <code>TestCase.scriptPath</code>, puis vers <code>TestCaseResponse.scriptPath</code>. Confirme également que le constructeur ou le mapper (<code>ResponseMapper</code>, <code>ModelMapper</code>, etc.) inclut ce champ. En cas de champ <code>transient</code> ou omis dans la construction de la réponse, corrige-le. **Conseil** : Active les logs SQL ou utilise un débogueur pour inspecter l'entité <code>TestCase</code> après sauvegarde, afin de confirmer que tous les champs attendus sont bien persistés et retournés dans la réponse – surtout les champs optionnels non annotés <code>@NonNull</code>.

Corrections automatiques par IA

L'IA a detecte des erreurs dans certains scripts generes et les a corriges automatiquement.

create project

3 correction(s)

FAILURE

```
=== Tentative 1/3 ===
Erreur détectée: [ERROR] COMPILATION ERROR :
[ERROR] /C:/tmp/ms-execution/exec-1126529147941670935/Backend/ms_gestion/src/test/java/suites/unit/ProjectServiceCreateHappyTest.java:[79,9]
cannot find symbol
[INFO] BUILD FAILURE
[ERROR] Failed to execute goal org.apache.maven.plugins:maven-compiler-plugin:3.14.1:testCompile (default-
testCompile) on project ms_gestion: Compilation failure
[ERROR] /C:/tmp/ms-execution/exec-1126529147941670935/Backend/ms_gestion/src/test/java/suites/unit/ProjectServiceCreateHappyTest
Résultat: Script corrigé (3451 chars), relance Maven...
Résultat: Encore en échec (exit=1).

=== Tentative 2/3 ===
Erreur détectée: [ERROR] COMPILATION ERROR :
[ERROR] /C:/tmp/ms-execution/exec-1126529147941670935/Backend/ms_gestion/src/test/java/suites/unit/ProjectServiceCreateHappyTest.java:[33,13]
cannot find symbol
[ERROR] /C:/tmp/ms-execution/exec-1126529147941670935/Backend/ms_gestion/src/test/java/suites/unit/ProjectServiceCreateHappyTest.java:[36,13]
cannot find symbol
[INFO] BUILD FAILURE
[ERROR] Failed to execute goal org.apache.maven.plugins:maven-compiler-plugin:3.14.1:testCompile (default-
testCompile) on projec
Résultat: Script corrigé (3516 chars), relance Maven...
Résultat: Encore en échec (exit=1).

=== Tentative 3/3 ===
Erreur détectée: [ERROR] COMPILATION ERROR :
[ERROR] /C:/tmp/ms-execution/exec-1126529147941670935/Backend/ms_gestion/src/test/java/suites/unit/ProjectServiceCreateHappyTest.java:[40,26]
cannot find symbol
[INFO] BUILD FAILURE
[ERROR] Failed to execute goal org.apache.maven.plugins:maven-compiler-plugin:3.14.1:testCompile (default-
testCompile) on project ms_gestion: Compilation failure
[ERROR] /C:/tmp/ms-execution/exec-1126529147941670935/Backend/ms_gestion/src/test/java/suites/unit/ProjectServiceCreateHappyTes
Résultat: Script corrigé (3451 chars), relance Maven...
Résultat: Encore en échec (exit=1).
```

create envirenemnt

1 correction(s)

SUCCESS

```
=== Tentative 1/3 ===
Erreur détectée: [ERROR] COMPILATION ERROR :
[ERROR] /C:/tmp/ms-execution/exec-8096209451775046421/Backend/ms_gestion/src/test/java/suites/integration/EnvironmentServiceIT.java:[63,29]
incompatible types: com.pfe.platform.ms_gestion.entity.ProjectMemberId cannot be converted to java.lang.Long
[INFO] BUILD FAILURE
[ERROR] Failed to execute goal org.apache.maven.plugins:maven-compiler-plugin:3.14.1:testCompile (default-
testCompile) on project ms_gestion: Compilation failure
[ERROR] /C:/tmp/ms-execution/exec-80962
Résultat: Script corrigé (4760 chars), relance Maven...
Résultat: SUCCÈS après correction !
```

List environnement

2 correction(s)

SUCCESS

```
=== Tentative 1/3 ===
```

```
Erreur détectée: [ERROR] Tests run: 1, Failures: 1, Errors: 0, Skipped: 0, Time elapsed: 12.74 s <<<
FAILURE! -- in suites.integration.EnvironmentServiceIT
[ERROR] suites.integration.EnvironmentServiceIT.test_list_happy_path -- Time elapsed: 0.174 s <<< FAILURE!
[ERROR] Failures:
[ERROR] EnvironmentServiceIT>AbstractTestNGSpringContextTests.run:153->test_list_happy_path:78 Runtime
Vous n'êtes pas membre de ce projet
[ERROR] Tests run: 1, Failures: 1, Errors: 0, Skipped: 0
[INFO] BUILD FAILURE
[ERROR] Failed
Résultat: Script corrigé (4209 chars), relance Maven...
Résultat: Encore en échec (exit=1).
```

=== Tentative 2/3 ===

```
Erreur détectée: [ERROR] COMPILATION ERROR :
[ERROR] /C:/tmp/ms-execution/exec-
6696813744712490012/Backend/ms_gestion/src/test/java/suites/integration/EnvironmentServiceIT.java:[60,22]
incompatible types: com.pfe.platform.ms_gestion.entity.ProjectMemberId cannot be converted to java.lang.Long
[ERROR] /C:/tmp/ms-execution/exec-
6696813744712490012/Backend/ms_gestion/src/test/java/suites/integration/EnvironmentServiceIT.java:[62,15]
cannot find symbol
[INFO] BUILD FAILURE
[ERROR] Failed to execute goal org.apache.
Résultat: Script corrigé (4146 chars), relance Maven...
Résultat: SUCCÈS après correction !
```

add test case

3 correction(s)

FAILURE

=== Tentative 1/3 ===

```
Erreur détectée: [ERROR] Tests run: 1, Failures: 1, Errors: 0, Skipped: 0, Time elapsed: 12.59 s <<<
FAILURE! -- in suites.integration.TestCaseServiceIT
[ERROR] suites.integration.TestCaseServiceIT.test_add_happy_path -- Time elapsed: 0.152 s <<< FAILURE!
[ERROR] Failures:
[ERROR] TestCaseServiceIT>AbstractTestNGSpringContextTests.run:153->test_add_happy_path:103 Runtime Vous
n'êtes pas membre de ce projet
[ERROR] Tests run: 1, Failures: 1, Errors: 0, Skipped: 0
[INFO] BUILD FAILURE
[ERROR] Failed to execut
Résultat: Script corrigé (6209 chars), relance Maven...
Résultat: Encore en échec (exit=1).
```

=== Tentative 2/3 ===

```
Erreur détectée: [ERROR] COMPILATION ERROR :
[ERROR] /C:/tmp/ms-execution/exec-
13892590320815266371/Backend/ms_gestion/src/test/java/suites/integration/TestCaseServiceIT.java:[75,22]
incompatible types: com.pfe.platform.ms_gestion.entity.ProjectMemberId cannot be converted to java.lang.Long
[INFO] BUILD FAILURE
[ERROR] Failed to execute goal org.apache.maven.plugins:maven-compiler-plugin:3.14.1:testCompile (default-
testCompile) on project ms_gestion: Compilation failure
[ERROR] /C:/tmp/ms-execution/exec-1389259
Résultat: Script corrigé (6141 chars), relance Maven...
Résultat: Encore en échec (exit=1).
```

=== Tentative 3/3 ===

```
Erreur détectée: [ERROR] Tests run: 1, Failures: 1, Errors: 0, Skipped: 0, Time elapsed: 12.95 s <<<
FAILURE! -- in suites.integration.TestCaseServiceIT
[ERROR] suites.integration.TestCaseServiceIT.test_add_happy_path -- Time elapsed: 0.214 s <<< FAILURE!
java.lang.AssertionError: expected [src/test/java/com/example/user/AddUserTest.java] but found [null]
[ERROR] Failures:
[ERROR] TestCaseServiceIT>AbstractTestNGSpringContextTests.run:153->test_add_happy_path:122 expected
[src/test/java/com/example/user/AddUs
Résultat: Script corrigé (6141 chars), relance Maven...
Résultat: Encore en échec (exit=1).
```

Metriques

Top 5 des tests les plus lents

Test	Suite	Duree
List environnement	Suite INTEGRATION	24.7 s
add test case	Suite INTEGRATION	24.4 s
create project	Suite UNIT	11.1 s
create envirenemnt	Suite INTEGRATION	11.0 s

Temps d'execution par type

Type	Duree moyenne
UNIT	11.1 s
INTEGRATION	20.0 s

Tests instables

Aucun test instable detecte

Recommandations

- create project:

****Cause**** : Les classes `ProjectAccessService`` (ligne 33) et `ProjectService`` (ligne 36) sont introuvables, ce qui indique qu'elles ne sont pas dans le classpath de compilation des tests (probablement absentes dans le projet ou mal structurées).

****Correction**** : Vérifie que `ProjectAccessService.java`` et `ProjectService.java`` existent bien dans le module source (`src/main/java/com/pfe/platform/ms_gestion/service/``), qu'elles sont déclarées comme `public``, et qu'aucun erreur de compilation dans les sources ne bloque la compilation des tests. Si le projet est modulaire, assure-toi qu'elles sont exportées/visibles dans `module-info.java`` ou incluses dans les dépendances de compilation.

****Conseil**** : Exécute `mvn clean compile`` avant `mvn test-compile`` pour t'assurer que les classes sources sont compilées avant les tests.

- add test case:

****Cause**** : La méthode `testCaseService.add(suiteId, request)`` renvoie un `TestCaseResponse`` dont le champ `scriptPath`` est `null``, alors que le test s'attend à ce qu'il soit égal à `"src/test/java/com/example/user/AddUserTest.java"`` (lignes 117–122 dans `test_add_happy_path``). Cela indique que le service ou la couche de persistance ne transfère pas correctement ce champ depuis la requête vers l'entité ni vers la réponse.

****Correction**** : Vérifie que `TestCaseService.add(...)`` mappe explicitement le champ `scriptPath`` de `CreateTestCaseRequest`` vers `TestCase.scriptPath``, puis vers `TestCaseResponse.scriptPath``. Confirme également que le constructeur ou le mapper (`ResponseMapper``, `ModelMapper``, etc.) inclut ce champ. En cas de champ `transient`` ou omis dans la construction de la réponse, corrige-le.

****Conseil**** : Active les logs SQL ou utilise un débogueur pour inspecter l'entité `TestCase`` après sauvegarde, afin de confirmer que tous les champs attendus sont bien persistés et retournés dans la réponse — surtout les champs optionnels non annotés `@NonNull``.

Annexe - Scripts generes par IA

create project

```
package suites.unit;

import com.pfe.platform.ms_gestion.dto.request.CreateProjectRequest;
import com.pfe.platform.ms_gestion.entity.Project;
import com.pfe.platform.ms_gestion.entity.ProjectMember;
import com.pfe.platform.ms_gestion.repository.ProjectMemberRepository;
import com.pfe.platform.ms_gestion.repository.ProjectRepository;
import com.pfe.platform.ms_gestion.security.SecurityUtils;
import com.pfe.platform.ms_gestion.service.ProjectAccessService;
import com.pfe.platform.ms_gestion.service.ProjectService;

import org.mockito.InjectMocks;
import org.mockito.Mock;
import org.mockito.MockedStatic;
import org.mockito.MockitoAnnotations;
import org.testng.annotations.AfterMethod;
import org.testng.annotations.BeforeMethod;
import org.testng.annotations.Test;

import java.util.Optional;

import static org.mockito.ArgumentMatchers.any;
import static org.mockito.Mockito.*;
import static org.testng.Assert.*;

public class ProjectServiceCreateHappyTest {

    @Mock
    private ProjectRepository projectRepository;

    @Mock
    private ProjectMemberRepository projectMemberRepository;

    @Mock
    private ProjectAccessService projectAccessService;

    @InjectMocks
    private ProjectService projectService;

    private AutoCloseable mocks;
    private MockedStatic<SecurityUtils> mockedSecurity;

    @BeforeMethod
    public void setUp() {
        mocks = MockitoAnnotations.openMocks(this);
        mockedSecurity = mockStatic(SecurityUtils.class);
        mockedSecurity.when(SecurityUtils::getCurrentUserId).thenReturn(1L);
    }

    @AfterMethod
    public void tearDown() throws Exception {
        mockedSecurity.close();
        mocks.close();
    }

    @Test
    public void test_create_happy_path() {
        // Given
        CreateProjectRequest request = new CreateProjectRequest();
        request.setName("Projet Alpha");
        request.setDescription("Développement d'une nouvelle application web");

        when(projectRepository.existsByName("Projet Alpha")).thenReturn(false);
```

```

Project savedProject = new Project();
savedProject.setId(1L);
savedProject.setName("Projet Alpha");
savedProject.setDescription("Développement d'une nouvelle application web");
savedProject.setCreatedBy(1L);
when(projectRepository.save(any(Project.class))).thenReturn(savedProject);

ProjectMember savedMember = new ProjectMember();
savedMember.setId(1L);
savedMember.setProject(savedProject);
savedMember.setUserId(1L);
when(projectMemberRepository.save(any(ProjectMember.class))).thenReturn(savedMember);

// When
ProjectResponse result = projectService.create(request);

// Then
assertNotNull(result);
assertEquals(result.getName(), "Projet Alpha");
assertEquals(result.getDescription(), "Développement d'une nouvelle application web");
assertEquals(result.getCreatedBy(), Long.valueOf(1L));
verify(projectRepository).existsByName("Projet Alpha");
verify(projectRepository).save(any(Project.class));
verify(projectMemberRepository).save(any(ProjectMember.class));
}
}

```

create envirenemnt

```

package suites.integration;

import com.pfe.platform.ms_gestion.dto.request.CreateEnvironmentRequest;
import com.pfe.platform.ms_gestion.dto.response.EnvironmentResponse;
import com.pfe.platform.ms_gestion.entity.Environment;
import com.pfe.platform.ms_gestion.entity.Project;
import com.pfe.platform.ms_gestion.entity.ProjectMember;
import com.pfe.platform.ms_gestion.entity.ProjectMemberId;
import com.pfe.platform.ms_gestion.repository.CampaignRepository;
import com.pfe.platform.ms_gestion.repository.EnvironmentRepository;
import com.pfe.platform.ms_gestion.repository.ProjectMemberRepository;
import com.pfe.platform.ms_gestion.repository.ProjectRepository;
import com.pfe.platform.ms_gestion.service.EnvironmentService;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.context.SpringBootTest;
import org.springframework.security.authentication.UsernamePasswordAuthenticationToken;
import org.springframework.security.core.context.SecurityContextHolder;
import org.springframework.security.core.context.SecurityContextHolderImpl;
import org.springframework.test.annotation.Rollback;
import org.springframework.test.context.ActiveProfiles;
import org.springframework.test.context.TestPropertySource;
import org.springframework.test.context.testng.AbstractTestNGSpringContextTests;
import org.springframework.transaction.annotation.Transactional;
import org.testng.annotations.AfterMethod;
import org.testng.annotations.BeforeMethod;
import org.testng.annotations.Test;

import java.util.Collections;

import static org.testng.Assert.*;

@SpringBootTest
@ActiveProfiles("test")
@Transactional
@Rollback(true)
@TestPropertySource(properties = {
    "spring.datasource.url=jdbc:h2:mem:testdb;DB_CLOSE_DELAY=-1;MODE=PostgreSQL",
    "spring.datasource.driverClassName=org.h2.Driver",

```

```
"spring.datasource.username=sa",
"spring.datasource.password=",
"spring.jpa.hibernate.ddl-auto=create-drop",
"spring.jpa.database-platform=org.hibernate.dialect.H2Dialect"
})
public class EnvironmentServiceIT extends AbstractTestNGSpringContextTests {

@Autowired private EnvironmentService environmentService;
@Autowired private ProjectRepository projectRepository;
@Autowired private ProjectMemberRepository projectMemberRepository;
@Autowired private EnvironmentRepository environmentRepository;
@Autowired private CampaignRepository campaignRepository;

private Long projectId;
private Long userId = 1L;

@BeforeMethod
public void setUp() {
// Insert prerequisite Project
Project project = new Project();
project.setName("AddProject");
project.setDescription("Test project");
project.setGitRepoUrl("https://github.com/org/webapp");
project.setCreatedBy(userId);
project = projectRepository.save(project);
projectId = project.getId();

// Insert ProjectMember to satisfy checkMembership
ProjectMember projectMember = new ProjectMember();
projectMember.setId(new ProjectMemberId(userId, projectId));
projectMember.setProject(project);
projectMemberRepository.save(projectMember);

// Set security context
SecurityContextHolder.setContext(new SecurityContextImpl(
new UsernamePasswordAuthenticationToken(userId, null, Collections.emptyList())));
}

@AfterMethod
public void tearDown() {
SecurityContextHolder.clearContext();
}

@Test
public void test_add_happy_path() {
// Given
CreateEnvironmentRequest request = new CreateEnvironmentRequest();
request.setName("AddProject");
request.setBaseUrlWeb("https://app.example.com");
request.setBaseUrlApi("https://api.example.com");
request.setGitRepoUrl("https://github.com/org/webapp");
request.setGitBranch("develop");
request.setDatabaseType("POSTGRESQL");

// When
EnvironmentResponse result = environmentService.add(projectId, request);

// Then
assertNotNull(result);
assertNotNull(result.getId());
assertEquals(result.getName(), "AddProject");
assertEquals(result.getBaseUrlWeb(), "https://app.example.com");
assertEquals(result.getBaseUrlApi(), "https://api.example.com");
assertEquals(result.getGitRepoUrl(), "https://github.com/org/webapp");
assertEquals(result.getGitBranch(), "develop");
assertEquals(result.getDatabaseType(), "POSTGRESQL");

// Verify persistence in DB
Environment savedEnv = environmentRepository.findById(result.getId()).orElse(null);
assertNotNull(savedEnv);
```

```
assertEquals(savedEnv.getName(), "AddProject");
assertEquals(savedEnv.getProject().getId(), projectId);
}
}
```

List environnement

```
package suites.integration;

import org.springframework.test.context.TestPropertySource;

import com.pfe.platform.ms_gestion.MsGestionApplication;
import com.pfe.platform.ms_gestion.dto.request.CreateEnvironmentRequest;
import com.pfe.platform.ms_gestion.dto.response.EnvironmentResponse;
import com.pfe.platform.ms_gestion.entity.Environment;
import com.pfe.platform.ms_gestion.entity.Project;
import com.pfe.platform.ms_gestion.repository.EnvironmentRepository;
import com.pfe.platform.ms_gestion.repository.ProjectRepository;
import com.pfe.platform.ms_gestion.service.EnvironmentService;
import com.pfe.platform.ms_gestion.service.ProjectAccessService;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.context.SpringBootTest;
import org.springframework.security.authentication.UsernamePasswordAuthenticationToken;
import org.springframework.security.core.context.SecurityContextHolder;
import org.springframework.security.core.context.SecurityContextImpl;
import org.springframework.test.annotation.Rollback;
import org.springframework.test.context.ActiveProfiles;
import org.springframework.test.context.testng.AbstractTestNGSpringContextTests;
import org.springframework.transaction.annotation.Transactional;
import org.testng.annotations.AfterMethod;
import org.testng.annotations.BeforeMethod;
import org.testng.annotations.Test;

import java.util.Collections;

import static org.testng.Assert.*;

@SpringBootTest(classes = MsGestionApplication.class)
@ActiveProfiles("test")
@Transactional
@Rollback(true)
public class EnvironmentServiceIT extends AbstractTestNGSpringContextTests {

    @Autowired private EnvironmentService environmentService;
    @Autowired private EnvironmentRepository environmentRepository;
    @Autowired private ProjectRepository projectRepository;

    private Long projectId;
    private Long userId;

    @BeforeMethod
    public void setUp() {
        userId = 1L;

        Project project = new Project();
        project.setName("project-alpha");
        project.setDescription("Test project");
        project.setGitRepoUrl("git@github.com:example/project-alpha.git");
        project.setCreatedBy(userId);
        project = projectRepository.save(project);
        projectId = project.getId();

        SecurityContextHolder.setContext(new SecurityContextImpl(
            new UsernamePasswordAuthenticationToken(userId, null, Collections.emptyList())));
    }

    @AfterMethod
```

TestPlatform

Rapport de campagne

```
public void tearDown() {
    SecurityContextHolder.clearContext();
}

@Test
public void test_list_happy_path() {
    // Given
    CreateEnvironmentRequest request = new CreateEnvironmentRequest();
    request.setName("project-alpha");
    request.setBaseUrlWeb("https://app.example.com");
    request.setBaseUrlApi("https://api.example.com");
    request.setGitRepoUrl("git@github.com:example/project-alpha.git");
    request.setGitBranch("develop");
    request.setDatabaseType("postgres");

    environmentService.add(projectId, request);

    // When
    java.util.List<EnvironmentResponse> result = environmentService.list(projectId);

    // Then
    assertNotNull(result);
    assertFalse(result.isEmpty());
    assertEquals(result.size(), 1);

    EnvironmentResponse env = result.get(0);
    assertNotNull(env);
    assertEquals(env.getName(), "project-alpha");
    assertEquals(env.getBaseUrlWeb(), "https://app.example.com");
    assertEquals(env.getBaseUrlApi(), "https://api.example.com");
    assertEquals(env.getGitRepoUrl(), "git@github.com:example/project-alpha.git");
    assertEquals(env.getGitBranch(), "develop");
    assertEquals(env.getDatabaseType(), "postgres");
}
}
```

add test case

```
package suites.integration;

import org.springframework.test.context.TestPropertySource;

import com.pfe.platform.ms_gestion.MsGestionApplication;
import com.pfe.platform.ms_gestion.dto.request.CreateTestCaseRequest;
import com.pfe.platform.ms_gestion.dto.response.TestCaseResponse;
import com.pfe.platform.ms_gestion.entity.Project;
import com.pfe.platform.ms_gestion.entity.TestSuite;
import com.pfe.platform.ms_gestion.entity.TestCase;
import com.pfe.platform.ms_gestion.repository.ProjectRepository;
import com.pfe.platform.ms_gestion.repository.TestSuiteRepository;
import com.pfe.platform.ms_gestion.repository.TestCaseRepository;
import com.pfe.platform.ms_gestion.service.ProjectAccessService;
import com.pfe.platform.ms_gestion.service.TestCaseService;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.context.SpringBootTest;
import org.springframework.security.authentication.UsernamePasswordAuthenticationToken;
import org.springframework.security.core.context.SecurityContextHolder;
import org.springframework.security.core.context.SecurityContextImpl;
import org.springframework.test.annotation.Rollback;
import org.springframework.test.context.ActiveProfiles;
import org.springframework.test.context.testng.AbstractTestNGSpringContextTests;
import org.springframework.transaction.annotation.Transactional;
import org.testng.annotations.AfterMethod;
import org.testng.annotations.BeforeMethod;
import org.testng.annotations.Test;

import java.util.Collections;
```

```
import static org.testng.Assert.*;

@SpringBootTest(classes = MsGestionApplication.class)
@ActiveProfiles("test")
@Transactional
@Rollback(true)
public class TestCaseServiceIT extends AbstractTestNGSpringContextTests {

    @Autowired private TestCaseService testCaseService;
    @Autowired private TestSuiteRepository testSuiteRepository;
    @Autowired private ProjectRepository projectRepository;
    @Autowired private TestCaseRepository testCaseRepository;

    private Long projectId;
    private Long suiteId;
    private Long userId = 1L;

    @BeforeMethod
    public void setUp() {
        // Create project
        Project project = new Project();
        project.setName("Test Project");
        project.setDescription("Project for test case");
        project.setGitRepoUrl(null); // AI mode
        project.setCreatedBy(userId);
        project = projectRepository.save(project);
        projectId = project.getId();

        // Create suite linked to project
        TestSuite suite = new TestSuite();
        suite.setName("Test Suite");
        suite.setDescription("Suite for test case");
        suite.setProject(project);
        suite.setGitRepoUrl(null);
        suite.setGitBranch(null);
        suite.setModulePath(null);
        suite = testSuiteRepository.save(suite);
        suiteId = suite.getId();

        // Set security context
        SecurityContextHolder.setContext(new SecurityContextImpl(
            new UsernamePasswordAuthenticationToken(userId, null, Collections.emptyList())));
    }

    @AfterMethod
    public void tearDown() {
        SecurityContextHolder.clearContext();
    }

    @Test
    public void test_add_happy_path() {
        // Given
        CreateTestCaseRequest request = new CreateTestCaseRequest();
        request.setTitle("AddUserOperation");
        request.setDescription("Test addition d'un utilisateur");
        request.setType("unit");
        request.setSpringProfile("test");
        request.setPriority(2);
        request.setRiskLevel("medium");
        request.setGitRepoUrl("https://git.example.com/user-service");
        request.setScriptPath("src/test/java/com/example/user/AddUserTest.java");
        request.setTestData("userId=123&name=JohnDoe");
        request.setTags("user,authentication,core");
        request.setMaxDurationSeconds(45);
        request.setDatabaseType("postgresql");
        request.setTargetClassName("UserService");
        request.setDescriptionAI("Generate a unit test for UserService.addUser");
        request.setGeneratedCode("public void testAddUser() { /* generated */ }");
    }
}
```

```
// When
TestCaseResponse result = testCaseService.add(suiteId, request);

// Then
assertNotNull(result);
assertNotNull(result.getId());
assertEquals(result.getTitle(), "AddUserOperation");
assertEquals(result.getDescription(), "Test addition d'un utilisateur");
assertEquals(result.getType(), "UNIT");
assertEquals(result.getSpringProfile(), "test");
assertEquals(result.getPriority(), Integer.valueOf(2));
assertEquals(result.getRiskLevel(), "MEDIUM");
assertEquals(result.getGitRepoUrl(), "https://git.example.com/user-service");
assertEquals(result.getScriptPath(), "src/test/java/com/example/user/AddUserTest.java");
assertEquals(result.getTestData(), "userId=123&name=JohnDoe");
assertEquals(result.getTags(), "user,authentication,core");
assertEquals(result.getMaxDurationSeconds(), Integer.valueOf(45));
assertEquals(result.getDatabaseType(), "postgresql");
assertEquals(result.getTargetClassName(), "UserService");
assertTrue(result.getGenerated());
assertNotNull(result.getGeneratedCode());
assertEquals(result.getSuiteId(), suiteId);

// Verify persistence
TestCase persisted = testCaseRepository.findById(result.getId()).orElse(null);
assertNotNull(persisted);
assertEquals(persisted.getTitle(), "AddUserOperation");
assertTrue(persisted.getGenerated());
assertEquals(persisted.getGeneratedCode(), "public void testAddUser() { /* generated */ }");
}
}
```